1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

Central hypothesis (expanded): In castration-resistant prostate cancer (CRPC), persistent AR signaling is maintained not only by systemic androgens but also via intratumoral de novo androgen synthesis and by distinct AR alterations (full-length AR amplification/mutation and AR splice variants such as AR-V7) that enable ligand-independent activity. These mechanisms operate through three core, high-confidence paths and a coordinated three-term pathway, collectively driving castration resistance even under standard ADT and AR-targeted therapies. This framework predicts that concurrent activation of intratumoral steroidogenesis and AR alterations synergistically sustains AR-driven transcription and tumor growth, while AR-V7 and GR-mediated bypass mechanisms contribute to variable therapeutic resistance. The paths are testable via integrated genomic, transcriptomic, proteomic, and metabolomic readouts in patient samples and preclinical models.

CORE PATH 1: Intratumoral de novo androgen synthesis fuels AR signaling via full-length AR when systemic androgens are suppressed. - Mechanistic definition (non-obvious nodes): - Intratumoral de novo androgen synthesis: local production of testosterone/DHT within tumor tissue independent of circulating hormones; relies on steroidogenic enzymes expressed in tumor cells or tumor-associated stroma. - Steroidogenic enzyme CNV/expression (CYP17A1, HSD3B1/2, AKR1C3, SRD5A1/2): gene copy-number gains or upregulated transcriptional activity that enable conversion of cholesterol-derived precursors to active androgens. - AR signaling activity: downstream transcriptional output of AR pathway (e.g., KLK3/PSA, TMPRSS2, NKX3-1) reflecting receptor activity. - Castration-resistant phenotype: continued proliferation and survival despite systemic castration levels of testosterone. - Node types: - Hormone level: intratumoral testosterone, dihydrotestosterone (DHT) - CNV/amplification: CYP17A1, HSD3B1, AKR1C3 amplification - Expression level: CYP17A1, HSD3B1/2, AKR1C3 transcripts - Phenotype: AR signaling activity score; CR phenotype - Evidence and confidence (examples and notes): - High confidence in concept of intratumoral androgen synthesis contributing to CRPC; multiple studies show steroidogenic enzyme upregulation in CRPC and intratumoral androgen presence. Evidence count: high for enzyme expression and local androgen production; PMIDs to be input where available. - Key supporting concepts: (a) intratumoral expression of CYP17A1 and AKR1C3 linked to androgen synthesis; (b) correlation between steroidogenic enzyme activity and AR target gene expression under androgen deprivation. - Example PMIDs: TBD - Mechanistic linkage: Upregulated steroidogenesis increases intratumoral androgens, sustaining full-length AR–driven transcription even when circulating androgens are suppressed, potentially interacting with AR amplification to amplify signal.

CORE PATH 2: AR-V7 splice variant expression co-occurs with AR amplification and enables ligand-independent AR signaling, driving resistance to AR-targeted therapies. - Mechanistic definition (non-obvious nodes): - AR-V7 splice variant: constitutively active, ligand-independent AR isoform lacking the ligand-binding domain; promotes transcription of AR target genes without androgens. - AR amplification: increased copy number of AR gene, boosting overall AR signaling capacity. - AR signaling activity: downstream transcriptional program driven by AR-V7 and/or full-length AR. - Castration-resistant phenotype: tumor growth despite AR-targeted therapy. - Node types: - Splice isoform expression: AR-V7 mRNA/protein levels - CNV/amplification: AR gene copy number - Expression level: AR full-length and AR-V7 transcripts - Phenotype: AR target gene expression, AR signaling activity, CR phenotype - Evidence and confidence (examples and notes): - High confidence that AR-V7 associates with resistance to enzalutamide/abiraterone in several CRPC cohorts; evidence supports a mechanism of ligand-independent AR activity. Evidence count: high; PMIDs: TBD - AR amplification commonly co-occurs with AR-V7 in resistant tumors, potentially amplifying AR-driven transcription. Evidence count: moderate–high; PMIDs: TBD - Mechanistic linkage: AR-V7 provides a ligand-independent transcriptional drive; when coupled with AR amplification, the threshold for AR signaling under low androgen conditions is further increased, promoting resistance to AR-targeted therapies.

CORE PATH 3: AR pathway bypass via glucocorticoid receptor (GR) activation and alternate signaling crosstalk maintains AR target gene programs under AR-targeted therapy. - Mechanistic definition (non-obvious nodes): - Glucocorticoid receptor (GR) activation: GR can bind a subset of AR target gene promoters/enhancers and sustain AR-like transcriptional programs in the presence of AR inhibitors. - Crosstalk with AR signaling: overlapping transcriptional networks allow continuation of key AR-regulated processes despite AR blockade. - Castration-resistant phenotype: persistence of tumor growth under AR-targeted therapy due to GR-mediated compensation. - Node types: - Receptor activity: GR expression/activation status - Expression level: GR target gene signatures - Phenotype: CR phenotype under AR-directed therapy - Evidence and confidence (examples and notes): - Moderate-to-supportive evidence that GR can compensate in some CRPC contexts; in preclinical models, GR activation can regulate AR target genes when AR is inhibited. Evidence count: moderate; PMIDs: TBD - Mechanistic linkage: When AR signaling is pharmacologically or genetically suppressed, GR can assume a compensatory role, potentially explaining partial resistance to AR inhibitors and guiding combinatorial strategies (e.g., GR antagonism).

THREE-TERM PATH (intratumoral androgen levels → AR alteration → castration resistance): - Definition: This integrated path ties together intratumoral androgen abundance (via de novo synthesis or retention), AR alterations (amplification, mutation, or splice variants), and the emergence of castration resistance. The path yields testable predictions about the order and combination of molecular events and their impact on CRPC progression. - Nodes and sequence: - Term 1: Intratumoral androgen level (hormone concentration within tumor) - Term 2: AR alteration (CNV/mutation/AR-V7 expression; includes full-length AR amplification or variant expression) - Term 3: Castration resistance phenotype (proliferation/survival under low systemic androgens and AR-targeted therapy) - Evidence and confidence: high-level integration

supported by literature showing that intratumoral androgens persist in CRPC, AR alterations correlate with therapy resistance, and combined instances predict poorer response. PMIDs: TBD - Mechanistic linkage: Local androgen production increases AR activation potential; concurrent AR alterations lower the activation threshold or enable ligand-independent signaling; together they drive resistant growth under ADT/AR-targeted therapies.

CASE A Hierarchical Path Triage (Path quality and prioritization) - Tier 1 (2–3 core mechanistic paths, direct, high-confidence): - CORE PATH 1: Intratumoral de novo androgen synthesis leading to sustained AR signaling under systemic androgen deprivation. - CORE PATH 2: AR-V7 splice variant expression with AR amplification enabling ligand-independent AR signaling and resistance to AR-targeted therapy. - Tier 2 (3–4 supporting paths, moderate confidence): - CORE PATH 3: GR-mediated bypass and AR signaling cross-talk maintaining AR target programs during AR inhibition. - THREE-TERM PATH: Integrated pathway linking intratumoral androgen levels to AR alterations and acquisition of castration resistance. - Tier 3 (exploratory paths): - Epigenetic remodeling that sustains AR target gene programs independent of AR ligand status. - Role of steroidogenic enzyme co-factors and tumor microenvironment in sustaining intratumoral androgen synthesis (e.g., stromal-tumor interactions). - Other AR co-regulators (e.g., FOXA1, HOXB13) that modulate AR activity in the CRPC context. - Synthesis: central mechanistic question — What is the relative contribution and interplay of intratumoral androgen synthesis, AR alterations (amplification/mutation/variants), and GR/crosstalk in sustaining AR signaling and driving castration resistance? Anchor genes/entities — AR (full-length and AR-V7), CYP17A1, AKR1C3, HSD3B1, GR; supporting entities — AR target genes (KLK3/PSA,TMPRSS2,NKX3-1), steroidogenic enzymes, and signaling partners (FOXA1, HOXB13). - Path-quality warnings: - Potential circular inference if AR activity readouts are used to infer upstream determinants; require orthogonal measurements (genomics, proteomics, metabolomics). - Heterogeneity across patient tumors may shift tier assignment; single-sample extrapolations should be avoided. - AR-V7 detection methods (RNA vs protein) may yield discordant results; careful assay selection required.

EXPERIMENTAL APPROACH (feasible methods and techniques to test the expanded hypothesis) - In silico and bioinformatic analyses: - Aggregate and reanalyze publicly available CRPC datasets (RNA-seq, WGS, proteomics) to quantify: AR copy number, AR-V7 expression, CYP17A1/HSD3B1/AKR1C3 expression, GR expression, and AR target gene signatures; correlate with clinical therapy responses and time to progression. - Single-cell RNA-seq to map co-expression of AR-V7, steroidogenic enzymes, and GR across tumor cells and stromal components. - In vitro studies: - Prostate cancer cell lines (e.g., LNCaP, 22Rv1) under androgen-deprived conditions; manipulate AKR1C3/CYP17A1 expression via overexpression or CRISPRa/CRISPRi; measure intratumoral androgen levels (LC-MS/MS), AR target gene expression, and proliferation under ADT/enzalutamide/abiraterone exposure. - Generate isogenic lines differing in AR status (AR amplification, AR-V7 expression) to assess combined effects on AR signaling and cell growth under low androgen conditions. - GR modulation experiments: use GR agonists/antagonists to test compensatory transcriptional programs and sensitivity to AR inhibitors. - Organoid and patient-derived xenograft (PDX) models: - Develop CRPC organoids or PDXs that recapitulate intratumoral steroidogenesis; quantify intratumoral androgens by LC-MS/MS; test responsiveness to AR-targeted therapies and GR modulators. - Proteomics and metabolomics: - Mass spectrometry-based quantification of intratumoral androgens (testosterone, DHT) and steroidogenic enzyme activity in tumor tissues. - Phosphoproteomics to map signaling adaptations downstream of AR variants and GR crosstalk. - Functional readouts and endpoints: - AR signaling activity through target gene expression; proliferation assays under androgen-deprived conditions; time-to-progression in model systems; resistance readouts to enzalutamide/abiraterone. - Timeline and integration: - Initial in silico analyses to map candidate determinants and interactions; followed by in vitro functional assays; culminate with organoid/PDX validation and therapeutic perturbation studies.

JUSTIFICATION (concise background and rationale for the expanded hypothesis) - CRPC remains driven by AR signaling despite systemic androgen deprivation. There is substantial evidence that CRPC tumors sustain intratumoral androgens via de novo synthesis and upregulation of steroidogenic enzymes (e.g., CYP17A1, AKR1C3, HSD3B1), enabling continued AR activity even when circulating testosterone is suppressed.1-3 AR-V7 splice variants confer ligand-independent activity and are associated with resistance to AR-targeted therapies, supporting a second high-confidence mechanistic path.4-6 Additional lines of evidence support GR-mediated bypass as a compensatory mechanism under AR inhibition, highlighting pathway redundancy in maintaining AR-like transcription programs.7-9 The proposed integrated model (three-term path and hierarchical triage) provides a concrete framework to dissect the relative contributions and interactions of these mechanisms and to identify combinatorial strategies (e.g., targeting intratumoral steroidogenesis, AR-V7, and GR) to overcome resistance. The experimental plan uses clinically relevant models (CRPC cell lines, organoids, PDXs) and multi-omics readouts to generate testable predictions and actionable targets.

Notes on references and evidence availability: The high-level concepts (intratumoral androgen synthesis, AR-V7–mediated resistance, and GR cross-talk) are well supported in the CRPC literature, though specific PMIDs should be appended once checked against the latest literature databases for precise citations.

2. Why This Gene — Not Others

3. Clinically Testable Predictions

Prediction 1: The interaction between the imaging feature and anchor gene predicts clinical outcome.

Prediction 2: Spatial analysis will confirm the mechanistic link at the tissue level.

Prediction 3: Functional perturbation will establish causality.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions